

Vamsi Priya Sakhamuri

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Profile

Motivated researcher with a strong foundation in applied mathematics, computational modeling, and artificial intelligence. Passionate about developing mathematically rigorous and computationally efficient methods to solve real-world and theoretical problems. Seeking to contribute to interdisciplinary research bridging mathematical theory and intelligent systems.

Education

Old Dominion University, VA, USA

Master's in Computer Science, May 2025

- Relevant Coursework: Algorithms and Data Structures, Intro to Data Science & Analytics, Intellig Interactive System, Data Analytics and Big Data, ML OPS For Generative AI
- Graduate Research Assistant, Department of Systems & Software Engineer

DVR & DR.HS MIC College of Technology, Andhra Pradesh, India

Bachelor's in Computer Science, May 2023

- Relevant Coursework: Linear Algebra & Differential Equations, Problem Solving Approaches, Vector Calculus and Fourier Transformations, Data Structures through C, Probability & Statistics, Advanced Data Structures, Discrete Mathematics, Design & Analysis of Algorithms, R Programming

Research & Publications Experience

Graduate Research Assistant, Old Dominion University – Norfolk, VA

- Collaborated with faculty on research projects involving data-driven modeling and predictive analytics to enhance student retention and learning outcomes.
- Designed and implemented ETL pipelines (SQL, Python) for large-scale institutional datasets; optimized data cleaning, feature extraction, and validation workflows.
- Applied statistical and machine learning methods—including regression, decision trees, and time series forecasting—to model behavioral and academic trends.
- Created analytical dashboards (Power BI, Tableau) to visualize key institutional metrics and communicate quantitative insights to academic leadership.
- Documented research architecture, metadata, and reproducibility protocols to support ongoing computational research in educational data systems.
- Investigated the mathematical basis of model performance and data integrity, deepening understanding of applied optimization and inference methods.
- Contributed to the development of scalable research infrastructure integrating platforms such as Kaltura and SharePoint for academic analytics.

Publication

- Developed optimization algorithms for virtual machine allocation and job scheduling in cloud computing, enhancing resource utilization and operational efficiency.
- Implemented heuristic and priority-based scheduling models using Python and CloudSim, evaluating performance under different cost, time, and throughput constraints.
- Analyzed performance trade-offs, providing insights into efficient cloud resource management strategies.
- Published research findings, contributing to advancements in cloud computing resource scheduling.

Independent Research Projects

- **Predictive Modeling of Housing Price Index Changes** – Applied regression models and RMSE validation to forecast market fluctuations.

- **Data Visualization of Airbnb Listings in Albany** – Utilized EDA and Tableau visualizations to identify trends and correlations in pricing.
- **Irrigation Data Analysis** – Performed time-series modeling to estimate water consumption and efficiency in agricultural systems.

Teaching Experience

- Mentored undergraduate students in data analytics, predictive modeling, and visualization projects, providing guidance on research design and implementation.
- Assisted faculty in developing and evaluating data science lab modules, assignments, and research-based coursework.
- Led a student-led initiative called Coding Club, organizing several coding competitions, workshops, and peer-learning sessions to promote collaborative problem-solving.
- Conducted technical workshops on Python, SQL, and machine learning fundamentals to enhance students' computational and analytical skills.
- Guided student teams in debugging, optimizing algorithms, and presenting research-driven coding solutions.

Technologies

Programming & Modeling: Python, R, SQL, Java, MATLAB

Mathematical Tools: Linear/Nonlinear Optimization, Regression, Probability Theory, Numerical Analysis

Data Science & Visualization: Power BI, Tableau, Pandas, NumPy, Scikit-learn

Databases: MySQL, SQL Server, SQLite

Cloud & ETL Tools: Informatica, Databricks, Microsoft Project, ServiceNow

Web Technologies: HTML, CSS, JavaScript, Django

Version Control: Git, GitHub

Professional Experience

Systems & Development Assistant, Old Dominion University - Norfolk, VA May 2025 - Present

- Maintained and updated academic course data in the NOC (Network Operations Center) portal, ensuring accuracy across instructional systems and scheduling tools.
- Supported video content management by uploading, organizing, and monitoring lecture recordings in Panopto, streamlining access for faculty and students.
- Collaborated with IT staff to troubleshoot system issues and provided technical support related to academic media and course data tools.

Data Insights Analyst, India May 2022 - May 2023

- Built and maintained interactive dashboards and reports using Power BI and Tableau to support strategic decision-making.
- Conducted statistical analysis and pattern recognition using Python to generate data-driven recommendations.
- Utilized predictive models such as Linear Regression and Decision Trees to forecast trends and stakeholder needs.
- Translated complex datasets into actionable insights, enabling informed decisions across departments.
- Collaborated with cross-functional teams to enhance data-driven strategies and improve business outcomes.

Quality Assurance Tester, DVR & DR.HS MIC College of Technology - India Dec 2021 - May 2022

- Conducted comprehensive quality assurance testing on website content and functionality to ensure optimal user experience.
- Verified and updated web content for accuracy, consistency, and proper display across various pages and devices.
- Ensured cross-browser and cross-device compatibility to improve accessibility and performance.
- Utilized SQL and Python (Pandas, NumPy) for data validation, processing, and error detection in large-scale datasets. Reported and tracked defects, provided feedback to developers, and participated in regression testing.